# Reducing Readmission Disparities in Hospitals Serving High Volumes of Patients With Limited English Proficiency

## *Evidence From New Jersey*

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**Background:** Despite national attention to address disparities in health care, significant language-based health inequities exist in the United States.

**Objectives:** To evaluate whether readmissions for patients with limited English proficiency (LEP) are associated with the LEP volume of the hospital and to determine whether and to what extent the effect of hospital LEP volume on readmissions for LEP patients is conditional on the hospitals' nurse work environment.

**Research Design:** Cross-sectional analysis using 3 data sources from 2016: New Jersey Discharge Data Collection System, RN4CAST-US survey, and American Hospital Association Annual Survey.

**Subjects:** A total of 424,745 patients, 9.2% of which were defined as having LEP (n = 38,906), in 68 hospitals.

**Measures:** The RN4CAST survey utilized the Practice Environment Scale of the Nursing Work Index to measure nurses' ratings of their hospitals' nurse work environment. The outcome variable was 7-day readmissions, representing a potentially avoidable re-visit to the hospital. Hospital LEP volume was measured as the percentage of index admissions of LEP patients.

**Results:** In the fully adjusted stratified model, in hospitals with poor work environments, a 10-percentage point increase in LEP volume was significantly associated with 6% higher odds of a 7-day readmission for LEP patients [OR = 1.06, 95% CI (1.04–1.08), $P < 0.001$]. In hospitals with favorable environments, a 10-percentage point increase in LEP volume was associated with 2% higher odds of a 7-day readmission; however, this difference was not statistically significant [OR = 1.02, 95% CI (0.99–1.06)].

**Conclusion:** Readmission disparities were greatest in hospitals serving higher proportions of patients with LEP; however, the disparities were rendered insignificant in hospitals with the favorable nurse work environments.

**Key Words:** minority serving hospital, nurse work environment, limited English proficiency, LEP-serving hospital, readmission disparities

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Despite national attention to address disparities in health care,[1] significant language-based health inequities exist in the United States (U.S.). For example, patients with limited English proficiency (LEP) experience increased 30-day hospital readmissions, longer lengths of stay, and decreased satisfaction with the health care they received during hospitalizations as compared with English-proficient patients.[2–4] A multitude of mechanisms may explain these differences such as unaddressed language barriers, discrimination, and poor access to care and high-quality insurance.[4] Emerging evidence also suggests that certain characteristics of hospitals, such as the adequacy of nursing resources, are associated with outcome disparities for patients with LEP, for example, in reducing 72-hour emergency department revisits for patients with LEP.[5] Nurses—the clinician most frequently communicating with hospitalized patients—are well-positioned to reduce disparities for patients with LEP by identifying and addressing patients' language and other complex social needs. Another important factor associated with outcome disparities among minoritized patients is whether the hospital they receive services in

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provides a disproportionate share of care to minority patients—known as minority-serving hospitals.

There has been an alarming lag in addressing the poor quality of care for patients in minority-serving hospitals[6,7]—defined variably across the literature but generally as hospitals serving a disproportionate share of minority patients (eg, racial/ethnic minorities and/or economically disadvantaged patients).[6,8] Hospitals serving a high proportion of racial minorities are more likely to have worse nurse environments (eg, settings characterized by insufficient staffing ratios, poor nurse-physician collaborations, and lack of managerial support) and worse nurse-reported patient outcomes, as compared with hospitals that serve lower proportions of minority patients. For example, one study found that nurses working in hospitals with higher concentrations of Black patients reported poorer confidence in patients' readiness for discharge, more frequent care complaints from patients and their family members, more hospital-acquired infections, and less satisfaction with care.[9] Patients with LEP are an understudied minority population that preliminary research suggests may similarly experience inequities depending on where they receive care.[10] However, research defining hospitals by the proportion of LEP patients they serve is sparse, limiting efforts to address disparities among this understudied population. While most studies define minority-serving by the number of racial and ethnic minority patients served, few consider the volumes of LEP patients as a minoritized population. This is an important gap in the literature to address, as the United States is growing in linguistic diversity.[11]

Within minority-serving hospitals, patient outcomes may be conditional based on the adequacy of hospital nursing resources, defined by the staffing ratios, skill mix, nurse educational levels, and nurse work environments within hospitals. Nurse work environments have been identified as an important modifiable lever to reduce patient care inequities, for example, in reducing readmissions for Hispanic surgical patients;[9,12] however, a critical research gap is understanding the role of nurse work environments in minority-serving hospitals on outcome disparities. Importantly, there are no studies defining minority-serving hospitals by the volume of LEP patients served.

Thus, the objectives of this study were to evaluate whether hospital readmissions for LEP patients are associated with the LEP-serving volume of the hospital and to determine whether and to what extent the effect of hospital LEP-serving volume on readmissions for LEP patients is conditional on the hospitals' nurse work environment. Readmissions are an important outcome to examine given they often signal gaps in care during the initial hospital stay or poor coordination during discharge and follow-up.[13] Better work environments may provide nurses with adequate time, managerial support, and resources to address LEP patient's medical and social needs, including the time needed to acquire and effectively utilize language resources.[10] We hypothesized LEP patients will have higher readmission rates in higher LEP-serving hospitals as compared with LEP patients in lower LEP-serving hospitals, but the effect of LEP patient volume on readmissions will be explained, at least in part, by differences in the nurse work environment.

## METHODS

### Study Design, Data, and Sample

This study used 3 linked data sources from patients, nurses, and hospitals in New Jersey in 2016: (1) New Jersey Discharge Data Collection System (NJDDCS), (2) RN4CAST-US nurse survey data, and (3) American Hospital Association (AHA) Annual Survey data. New Jersey is an ideal state to examine LEP outcome disparities, given that approximately 40% of New Jersey residents speak a language other than English, making New Jersey among the most linguistically diverse states in the United States.[14] In addition, New Jersey's state-level discharge data collection systems are unique among other states, since the NJDDCS includes information about patient's preferred language, which provides an opportunity to study relationships between patient outcomes, the LEP volume of the hospital, and hospitals' nurse work environment.

The NJDDCS is managed by the New Jersey Department of Health and collects inpatient hospital discharge records from 100% of patients in all New Jersey hospitals and demographic data used in this study.

The RN4CAST-US is a survey of registered nurses that used a modified Dillman approach[15] by researchers at the Center for Health Outcomes and Policy Research to sample nurses used in hospitals. Mailed surveys were sent to the home addresses of a 30% random sample of actively licensed registered nurses in 4 large, geographically diverse states: California, Florida, New Jersey, and Pennsylvania (n = 231,000). The RN4CAST survey had a response rate of 26%. No statistically significant differences were found on the majority of survey measures between respondents and nonrespondents. Nurses were asked to provide information on their perceived work environments. Importantly, nurses used in hospitals were asked to report the name of their hospital employer, which allows for aggregation of survey response items.[16] This data source provided a measure of the hospital nurse work environment. The RN4CAST survey received Institutional Review Board (IRB) approval from the University of Pennsylvania.

The American Hospital Association Annual Hospital Survey provided information about hospital characteristics used for model risk adjustment such as teaching status, bed size, and technology status.

### Study Sample

This study evaluated a sample of hospitalized adults discharged from 1 of 68 New Jersey hospitals for a medical or surgical diagnosis between January 1, 2016, and December 31, 2016. Patients were included in the study sample if they were between the ages of 18 and 100 years old and were discharged from one of the NJ


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study hospitals. Patients admitted for psychiatric reasons, drug or alcohol, and labor and delivery-related conditions were excluded, as these populations have unique communication needs. Patients with a reported language preference other than English were defined as having LEP.[17]

Hospitals were included if they were adult nonfederal acute care hospitals located in New Jersey and had at least 3 nurse respondents from the RN4CAST survey reporting about the work environment. We determined that 3 nurse informants per hospital was sufficient to aggregate responses for the work environment given the intraclass correlation coefficient (ICC) of 0.7, which is considered strong.[18] This decision was also supported by previous studies using 2016 RN4CAST data which have used similar nurse respondent minimums.[19,20] There was an average of 37 nurses per hospital in our sample (range: 3–129 nurse respondents per hospital).

The final analytic sample for this study included 424,745 patients, 9.2% of which were defined as having LEP (n = 38,906), cared for in 1 of the 68 hospitals in our sample. This is a representative sample of NJ hospitals since the final sample represented 94.4% of general acute care hospitals in NJ.

## Measures

### Nurse Work Environment

The RN4CAST survey utilized the Practice Environment Scale of the Nursing Work Index (PES-NWI) to measure nurses' perceived ratings of their hospitals' nurse work environment. The PES-NWI is a 31-item tool with questions related to 5 subscales: (1) nurse participation in hospital affairs, (2) nursing foundations for quality of care, (3) nurse manager ability, (4) staffing and resource adequacy, and (5) collegial nurse-physician relations.[21] The survey items are measured on a 4-point Likert scale from strongly agree to strongly disagree. Reliability generalization has been demonstrated for this scale, which has been endorsed by the National Quality Forum.[22] A meta-analysis demonstrated a mean score reliability of 0.922 ($P < 0.05$), based on 38 studies (n = 68,278).[23] Nurse responses were aggregated for each subscale to create a hospital-level composite measure based on the average of the 5 subscale scores.[21] This hospital-level aggregation of PES-NWI scores is a common methodology that has been performed for decades of research examining hospital nurse work environments.[19,24,25] Each hospital-level nurse work environment score was then categorized as "poor" if the hospital score was in the bottom 25th percentile, or "favorable" if in the top 75th percentile.[19,26]

### 7-Day Readmission

The outcome variable for this study was 7-day readmissions, provided by the NJDDCS dataset. A 7-day readmission was defined as a re-hospitalization to any NJ study hospital that occurred within 7 days of the discharge date from the index hospitalization. This outcome was selected as it represents an acute, potentially avoidable re-visit to the hospital due to the short period of time between the initial hospitalization and readmission.[13] While not all readmissions can be prevented, there is responsibility on the hospital to adequately address patient needs and provide appropriate discharge planning to avoid readmissions.[27]

### LEP Volume

Hospital LEP volume was measured as a continuous variable as the percentage of index admissions (defined as the first hospitalization occurring in the study period) of LEP patients.

### Covariates

Patient covariates included for model risk adjustment were age, sex, Charlson Comorbidity Index (CCI), diagnostic-related groups (DRG), race, ethnicity, and insurance status and were chosen based on prior literature examining outcomes of patients with LEP.[28,29] Hospital covariates were measured through the AHA survey data and include bed size, teaching status, and technology status. Bed size is defined as the number of beds a hospital has and was categorized as small < = 100 beds, medium = < = 100–250 beds and large = > 250 beds. Teaching status is based on the ratio of medical residents to fellows per bed. Nonteaching hospitals have no medical residents/fellows. Minor teaching hospitals have a ratio that is < 1:4. Major teaching hospitals have a ratio that is > = 1:4. Technology status is defined by a hospital's ability to perform open-heart surgery or major organ transplantation. High-technology hospitals have the capacity to perform such procedures and non–high-technology hospitals do not.

## Statistical Analyses

Descriptive statistics (percentages, means, SDs) were used to analyze the distribution of patients across study hospitals. Differences in hospital characteristics (nurse work environment, bed size, teaching status, and technology status) among individuals with and without LEP were assessed using *t* tests for continuous variables and chi-square tests for categorical variables.

Sequential multivariable logistic regression models examining the association between hospital LEP volume and odds of a 7-day readmission for LEP patients were performed in a series of unadjusted and adjusted models. In addition, stratified models were performed to examine differences in the relationship between hospital LEP-serving volume and readmission outcomes by nurse work environment. Specifically, the sample was divided into 2 groups based on nurse work environment quality (eg, favorable vs. poor), allowing for separate analyses within each group to determine whether the association between LEP-serving volume and readmission variated by nurse work environment. Standard errors were estimated accounting for clustering using the Huber-White formulation. All analyses were conducted using Stata 17.[30]

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## RESULTS

Patients with LEP were more likely to receive care in hospitals characterized by poor nurse work environments (25.0% vs. 18.4%, $P < 0.001$) (Fig. 1), > = 250 beds (77.3% vs. 76.0%, $P < 0.001$), major teaching status (27.8% vs. 24.4%, $P < 0.001$), and high technology capabilities (39.9% vs. 37.9%, $P < 0.001$) (Table 1). Hospital LEP-serving volume ranged from 1.7% to 87.9%. Hospitals serving a higher volume of LEP patients had higher 7-day readmissions on average, though the correlation was considered weak (r = 0.38) (Fig. 2). The full table of patient demographic characteristics (Supplemental Table 1, Supplemental Digital Content 1, http://links.lww.com/MLR/D110) and hospital characteristics (Supplemental Table 2, Supplemental Digital Content 1, http://links.lww.com/MLR/D110) are provided in the Appendix, Supplemental Digital Content 1, http://links.lww.com/MLR/D110.

In the unadjusted model, every 10-percentage point increase in hospital LEP volume was associated with LEP patients having 5% higher odds of a 7-day readmission [OR = 1.05, 95% CI (1.03–1.07), $P < 0.001$] (Table 2). Adjusting for patient and hospital characteristics, a 10-percentage point increase in hospital LEP volume was associated with 6% higher odds of a 7-day readmission [OR = 1.06, 95% CI (1.03–1.09), $P < 0.001$]. When accounting for the nurse work environment, the odds of a 7-day readmission were slightly attenuated [OR = 1.05, 95% CI (1.02–1.08), $P < 0.05$], but remained statistically significant. In the fully adjusted model, favorable work

**TABLE 1.** Distribution of Patient Sample Across Hospital Characteristics

| Characteristic | LEP (n = 38,906) | Non-LEP (n = 385,839) | P |
|---|---|---|---|
| Work environment, n (%) | | | < 0.001 |
| Poor | 9710 (25.0) | 70,929 (18.4) | |
| Favorable | 29,196 (75.0) | 314,910 (81.6) | |
| Bed size, n (%) | | | < 0.001 |
| < = 100 beds | 130 (0.3) | 4360 (1.1) | |
| 101–250 beds | 8686 (22.3) | 88,240 (22.9) | |
| > = 250 Beds | 30,090 (77.3) | 293,239 (76.0) | |
| Teaching status*, n (%) | | | < 0.001 |
| Nonteaching | 8313 (21.4) | 133,094 (34.5) | |
| Minor teaching | 19,775 (50.8) | 158,762 (41.2) | |
| Major teaching | 10,818 (27.8) | 93,983 (24.4) | |
| Technology status†, n (%) | | | < 0.001 |
| Not high technology | 23,379 (60.1) | 239,440 (62.1) | |
| High technology | 15,527 (39.9) | 146,399 (37.9) | |

Notes. LEP = limited English proficiency; hospitals were categorized as having "poor" work environments if the aggregate hospital nurse work environment score was in the bottom 25th percentile, or "favorable" if in the top 75th percentile;

* = nonteaching: no medical residents/fellows, minor teaching: ratio of < 1:4 medical residents/fellows, major teaching: ratio of > = 1:4 medical residents/fellows;

† = not high-tech: do not have the ability to perform open-heart surgery or major organ transplantation, high-tech: have the capacity to perform open-heart surgery or major organ transplantation. Percentages may not total 100 due to rounding.

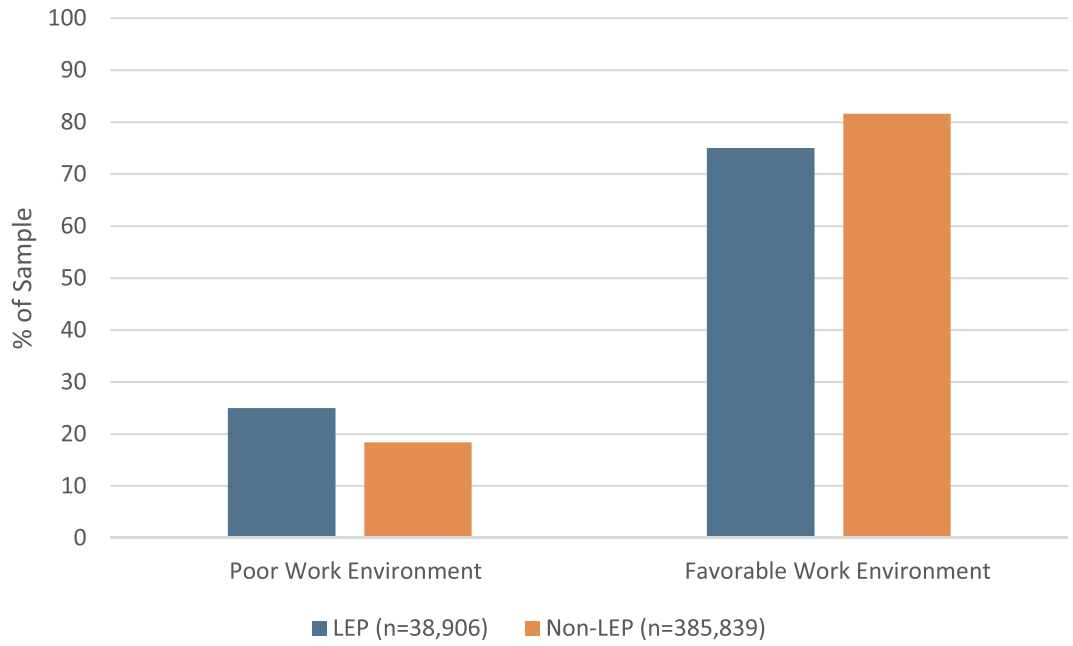


**FIGURE 1.** Distribution of patient sample across hospital nurse work environment. This bar graph demonstrates the distribution of the patient sample across nurse work environments, comparing the proportion of LEP patients and non-LEP patients in hospitals with poor versus favorable work environments. Patients with LEP are more likely to receive care in hospitals with poor work environments, as compared with patients without LEP ($P < 0.001$). full color online


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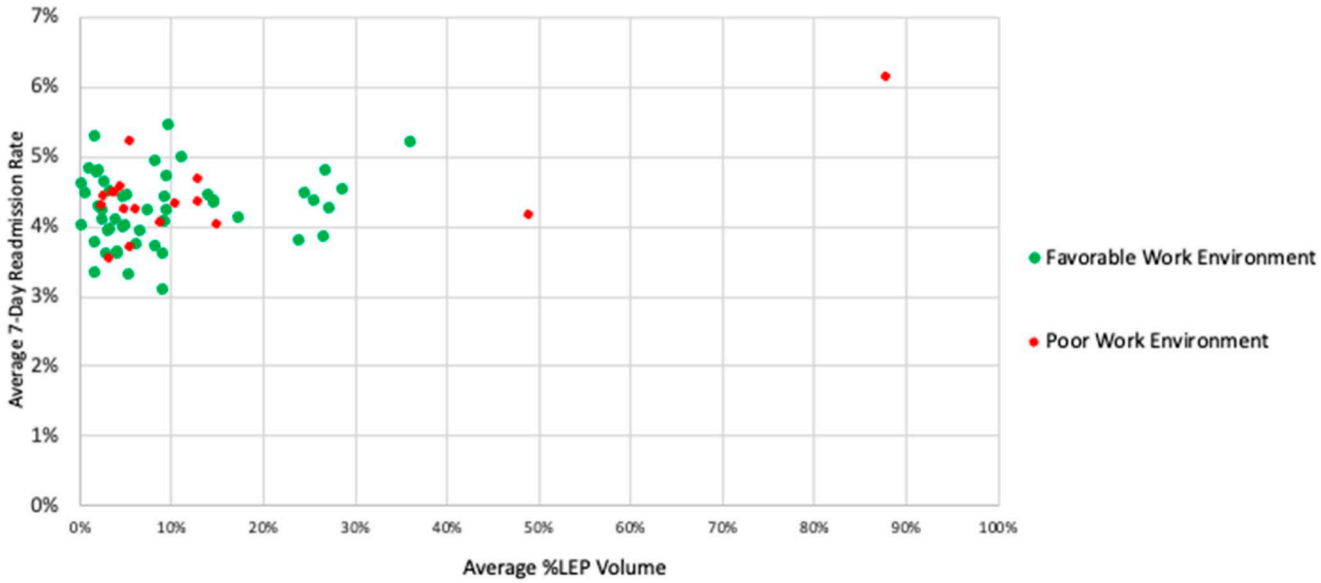


**FIGURE 2.** Scatter plot between 7-day readmission rates and hospital volume of LEP patients. This figure is a scatter plot demonstrating the relationship between hospital volume of LEP patients and average 7-day readmission rates. Hospital LEP volume ranged from 1.7% to 87.9%. Hospitals serving a higher volume of LEP patients had higher 7-day readmissions on average, though the correlation was considered weak (r = 0.38). full color online

environments were associated with 11% lower odds of a 7-day readmission, as compared with poor work environments, although the relationship did not reach statistical significance [OR = 0.89, 95% CI (0.77–1.03)]. All model covariate coefficients are provide in Supplemental Table 3 in the Appendix, Supplemental Digital Content 1, http://links.lww.com/MLR/D110.

To understand whether the nurse work environment moderates the effect of LEP volume on LEP odds of readmission, we ran stratified models (Table 2), comparing the effects of LEP volume on 7-day readmissions in poor work environments, as compared with favorable environments. Among hospitals with poor work environments, a 10-percentage point increase in hospital LEP volume was significantly associated with 6% higher odds of a 7-day readmission for LEP patients [OR = 1.06, 95% CI (1.04–1.08), $P < 0.001$]. Among hospitals with favorable environments, a 10-percentage point increase in hospital LEP volume was associated with 2% higher odds of a 7-day readmission; however, this difference was not

**TABLE 2.** Sequential and Stratified Models Indicating the Association Between Hospital LEP Volume and Readmission Outcomes of LEP Patients, and the Role of the Nurse Work Environment

| 7-day Readmission | Unadjusted OR (95% CI) | Adjusted for hospital1 and patient characteristics2 OR (95% CI) | Adjusted for patient, hospital characteristics, and nurse work environment OR (95% CI) | Stratified model adjusted for hospital and patient characteristics2 OR (95% CI) |
|---|---|---|---|---|
| Sequential models | | | | |
| Percent LEP3 | 1.05** (1.03–1.07) | 1.06** (1.03–1.09) | 1.05* (1.02–1.08) | |
| Nurse work environment4 | | | 0.89 (0.77–1.03) | |
| Stratified models | | | | |
| Poor work environments | | | | |
| Percent LEP3 | | | | 1.06** (1.04–1.08) |
| Favorable work environments | | | | |
| Percent LEP3 | | | | 1.02 (0.99–1.06) |

**Notes.** OR indicates Odds Ratio;

1 = Hospital characteristics include bed size, teaching status, and technology status;
2 = Patient characteristics include age, sex, Charlson Comorbidity Index, dummy variable for DRG, race, ethnicity, and insurance;
3 = Hospital-level measure of the % LEP index admissions, where a 1-unit increase represents a 10% increase in hospital % LEP volume;
4 = Nurse work environment is a binary variable comparing favorable work environments to poor work environment;
*$P < 0.05$,
**$P < 0.001$; logistic regression accounted for clustering of patients within hospitals.

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**TABLE 3.** Alternative Increases in Total Hospital LEP Volume on Odds of a 7-day Readmission

| Increase in hospital LEP volume | Calculation | Odds of a 7-day readmission |
|---|---|---|
| 10% | $1.06^1$ | 1.06 |
| 20% | $1.06^2$ | 1.12 |
| 30% | $1.06^3$ | 1.19 |
| 40% | $1.06^4$ | 1.26 |
| 50% | $1.06^5$ | 1.34 |

Note. The calculation column represents a multiplier effect per 10% increase in volume and the third column is the results of the calculations.

statistically significant [OR = 1.02, 95% CI (0.99–1.06)].

We then calculated the effect of a 20%, 30%, 40%, and 50% increase in total LEP volume on odds of a 7-day readmission, further showcasing the negative association of increased hospital LEP volume on LEP outcomes (Table 3). Every 20-percentage point increase in hospital LEP volume was associated with LEP patients having 12% higher odds of a 7-day readmission. Every 50-percentage point increase in total LEP volume was associated with 34% higher odds of a 7-day readmission. These differences demonstrate that higher LEP volume hospitals were associated with higher odds of readmission for LEP patients.

## DISCUSSION

Our study demonstrates a link between high LEP volume hospitals, perceived nurse work environments, and LEP readmission outcomes, such that higher LEP volume hospitals were associated with higher odds of a 7-day readmission for LEP patients, compared with non-LEP patients, although this relationship was only statistically significant in hospitals with the least favorable nurse work environments.

The findings of this study are consistent with other disparities research, which found worse quality of care and lower patient satisfaction within high-Black serving hospitals; however, the effects associated with the hospital-level volume of Black patients were explained in part by hospital nursing resources, such that adjusting for the nurse work environment or staffing markedly attenuated the effect associated with the hospital volume of Black patients on quality of care and patient satisfaction.[9] In addition, one study by Viscardi et al[31] (2022) found that adjusting for the nurse work environment attenuated the negative outcomes at hospitals serving higher proportions of patients who are economically disadvantaged. Thus, our findings extend the body of literature that characterizes relationships between hospital nurse work environments and patient outcomes in hospitals where disadvantaged populations receive care. Importantly, it identifies the potential advantage for LEP patient outcomes that could come with improving hospital nurse work environments.

Although our findings do not inform specifically about what the nurse work environment contributes to the association with reduced readmission disparities, a possible explanation for our findings is that nurses working in positive and well-resourced work environments may be better equipped to address unique LEP patient needs (ie, easily accessible and available language services, make connections to community-based resources, implement protocols, build trusting relationships, etc.) than nurses working in hospitals with negative work environments and minimal resources.[32–34] For example, hospitals with work environments that engage nurses in shared governance practices to influence organizational decision-making (ie, having the opportunity to serve on committees, participate in policy decisions, and consult with administration who listens to and responds to employee concerns)[21] may also prove beneficial for LEP patient care delivery. Shared governance has shown to be associated with improved nurse and patient outcomes.[35,36]

It is also essential to consider the quality and availability of interpreter services when addressing care for patients with limited English proficiency (LEP). Even in hospitals with favorable nurse work environments, inadequate interpreter services (eg, limited availability or technological barriers) can compromise care for LEP patients. Therefore, efforts to improve care delivery for this population should address both the conditions under which nurses provide care and the effectiveness of communication support. Notably, in-person medical interpreters are widely regarded as the gold standard, and health care institutions should prioritize their use over other modalities whenever feasible.[37,38]

Since 30-day readmissions are a national quality outcome measure supported by both Medicare and the Affordable Care Act (Hospital Readmission Reduction Act), this study (which examined 7-d readmissions and thus count as part of this quality measure) provides evidence that an organizational intervention that improves nurses' work environments could have the indirect effect of improving outcomes for LEP patients. Magnet recognition is the most well-known intervention to recognize hospitals with superior work environments for nurses. Recently, the Magnet Recognition Program has also recently integrated structures and processes related to diversity, equity, and inclusion (DEI) within the 2023 Magnet application. Examples include demonstrating instances of nurses delivering culturally sensitive care and policies or evidence of organizational advocacy initiatives for staff, patients, and families pertinent to DEI.[39]

### Limitations

The observational study design does not allow for determination of causality. However, longitudinal studies using RN4CAST data have exhibited similar results to studies utilizing a cross-sectional design.[40,41] Although this study used data from before the Covid-19 pandemic, the age of the data is likely not a limitation since research shows much of the progress made around patient safety and quality of care in the past twenty years has been reversed following the Covid-19 pandemic.[42] Therefore, it is plausible that disparities identified in our study persist


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today and may even be worse following the effects of the pandemic, which left many hospitals and clinicians burned out and under-resourced.[44] Although the PES-NWI has been demonstrated to show acceptable psychometric properties, the measure was not evaluated in these analyses, and it is assumed that the measure performed in a fashion similar to other studies. In addition, the use of data from 2016 was necessary due to the availability of the nurse survey data; however, despite its age, the dataset allowed us to examine unique variables (nurse work environment and outcomes for patients with LEP) in a large sample of New Jersey hospitals using one of the few datasets that includes information on patient language preference.

In addition, the study definition of individuals with LEP (those with a preferred language other than English) has limitations. It is possible that this definition does not fully capture the population, as some individuals may indicate English as their preferred language even when it is not in fear of receiving suboptimal care, or due to administrative errors when documenting this information. A preferred non-English language does not confirm limited English proficiency; however, this currently remains the most common definition of patients with LEP, as evidenced by previous studies.[29] It is also possible that language discordance can occur between English-proficient patients, and limited English-proficient clinicians, although this was not the focus of this study. Despite these limitations, this study provides novel evidence for actionable interventions to reduce readmission disparities for individuals with LEP.

## CONCLUSION

This study provides some of the first comparative evidence that some hospitals are better able to manage the needs of LEP patients and that the explanation for these differences may be tied to the quality of nurses' work environments. Future research should replicate the approach with similar and more recent data to determine the stability of the findings and discern the effects of the COVID-19 pandemic.

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